INCH-POUND

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SUPERSEDING
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30 November 2004

DETAIL SPECIFICATION

DISINFECTANT, FOOD SERVICE

Reactivated after 30 November 2004 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of disinfectant, based on trichloromelamine and potassium iodide, for use in food service preparation and disinfection of mess gear and eating utensils (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at http://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-121 - Barrier Materials, Greaseproof, Waterproof, Flexible, Heat-Sealable.

(Copies of these documents are available online at http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other government publications. The following other government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 162 - State Registration of Pesticide Products.

(Copies of these documents are available online at http://www.access.gpo.gov/ or from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

ASTM INTERNATIONAL

ASTM E 11 - Standard Specification for Wire Cloth and Sieves for Testing Purposes (DoD adopted).

(Copies of these documents are available online at http://www.astm.org/ or from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959.)

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

USP-NF - U.S. Pharmacopeia-National Formulary.

(Copies of this document are available online at http://www.usp.org/ or from the United States Pharmacopeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

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2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The food service disinfectant consists of trichloromelamine (see 6.4.1), a surface active material, anhydrous citric acid (see 6.4.1.1) conforming to the requirements of the U.S. Pharmocopeia-National Formulary (USP-NF), anhydrous monosodium dihydrogen phosphate (see 6.4.1.2), and potassium iodide in granular form in accordance with the USP-NF.

3.1.2 Surface active material. The surface active material shall be white to a light cream in color, anionic type. The active ingredient consists of linear sodium alkyl (C_{11}-C_{12}) aryl sulfonate (LAS).

3.1.2.1 Iodine uptake. The iodine uptake shall be no more than 4.5 milligrams (mg) of iodine per gram of surface active material when tested as specified in 4.5.1.1.

3.1.2.2 Active ingredient. The surface active material shall have an active ingredient content of not less than 81 percent when tested as specified in 4.5.1.2.

3.1.2.3 Moisture content. The moisture content of the surface active material shall be no greater than 1.25 percent when tested as specified in 4.5.1.3.

3.1.3 Potassium iodide. The potassium iodide shall be in granular form and shall conform to the requirements of the USP-NF.

3.2 Composition. When tested as specified in 4.4.2, the food service disinfectant shall conform to the formula specified in table I. The disinfectant requires the addition of a potassium iodide pouch, packaged separately.

TABLE I. Disinfectant formula.

Ingredient	Percent by weight	
	Minimum	Maximum
Trichloromelamine	26.9	27.9
Anionic linear sodium alkyl aryl sulfonate surface active material, technical (LAS),	11.5	12.5
Citric acid, anhydrous	51.0	55.0
Monosodium dihydrogen phosphate, anhydrous	Remainder	Remainder

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3.3 Pouches. The finished disinfectant product shall consist of a pouch labeled "A" and a pouch labeled "B". Unless otherwise specified (see 6.2), pouch A shall contain 3.43 ± 0.10 ounces of the disinfectant formula (see table I) and pouch B shall contain 1.34 ± 0.10 ounces of potassium iodide.

3.3.1 Pouch construction. The pouches, prepared as separate units, shall be constructed of heat sealable barrier material consisting of the following lamination of materials (from inside to outside surface): a 0.0015-inch polyethylene layer followed by a 0.002-inch aluminum foil layer followed by a 0.0015-inch polyethylene layer followed by a 25-pound basis weight pouch paper with 2-pound basis weight pouch paper with 2-pound basis weight per ream (24 by 36-500) lacquer coating. Pouches shall be one- or two-piece construction.

3.3.2 Pouch seams and closure. The side, top, and bottom seams of the two-piece pouch shall be heat sealed with the seams a minimum of 3/16 inch in width. In addition to the minimum of 3/16-inch seal on the top, a lip of at least 3/8 inch shall be present at the top of each pouch for joining and heat sealing both pouch A and B together. After each pouch has been filled and closure heat sealed, pouch A and pouch B shall be heat sealed together by joining the top 3/8-inch lip of each pouch.

3.3.3 Tearing notch. A tearing notch shall be positioned on the top seam of both pouches. The notch in each pouch shall be located in the top seam so that both pouches may be opened in one operation to ensure delivery of the contents during use.

3.3.4 Finished pouches. When tested as specified in 4.5.3.1, the finished pouches shall show no visible signs of deterioration. The exterior of the pouches shall be free from visible dust. The joined pouches shall be enclosed in a close fitting heat sealed bag fabricated from material conforming to type II of MIL-PRF-121.

3.3.5 Leakage of seams and closure of pouch. The seams and closure of the pouch shall be complete and continuous, shall not show signs of leakage or bubbling when immersed in water, and shall not allow the contents to get wet when tested as specified in 4.5.3.2.

3.4 Disinfectant finished product.

3.4.1 Titratable chlorine. The contents of the pouch A shall contain not less than 24.0 percent nor more than 26.5 percent of titratable chlorine when tested as specified in 4.5.2.1.

3.4.2 Acidity. The contents of the pouch A shall have a pH value not less than 2.5 and shall have a titratable acidity equivalent to not less than 200 mg of sodium hydroxide per gram of product when tested as specified in 4.5.2.2.

3.4.3 Surface active ingredient. The contents of the pouch A shall have not less than 9.1 percent nor more than 10.7 percent of active ingredient when tested as specified in 4.5.2.3. These values are slightly less than theoretical for there is seldom complete conversion in the test method used in 4.5.2.3.

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3.4.4 Solubility. The contents of the pouch A shall be soluble when tested as specified in 4.5.2.4. The titratable chlorine in the sample shall not be less than 152 mg per liter.

3.4.5 Moisture. The content of the pouch A shall have a moisture content of not more than 0.60 percent when tested as specified in 4.5.2.5.

3.4.6 Particle size. Not less than 95 percent, by weight, of the contents of the pouch A shall pass through a No. 20 standard sieve, and not less than 100 percent through a No. 10 standard sieve, both sieves conforming to ASTM E 11, when tested as specified in 4.5.2.6.

3.4.7 Bactericidal performance. When tested as specified in 4.5.2.7, the contents of pouch A shall give 99.999 percent kill in 90 seconds for at least two of the three tests.

3.4.8 Formulation stability. The contents of pouch A shall not contain less than 18 percent titratable chlorine when determined as specified in 4.5.2.8.

3.4.9 Surface tension. A solution of the contents of the pouch A shall have a surface tension of not more than 40 dynes per centimeter (cm) when prepared and tested as specified in 4.5.2.9.

3.5 Labeling. The labeling shall comply with the guidelines established under 40 CFR 162 for state registration of pesticide. The contractor shall use EPA registration number 40510-1 assigned to the US Army Research, Development and Engineering Command, Natick Soldier Center, AMSRD-NSC-CF-F, 15 Kansas St., Natick, MA 01760-5018. The contractor shall use the EPA establishment number assigned to the contractor's manufacturing plant.

3.5.1 Labels. The EPA registration number is 40510-1. The EPA approved labels shall be as shown in figures 1 and 2. The information found in figures 1 and 2 shall be legibly marked on each exterior bag containing the double-compartmented pouch and on each interior box.

3.5.2 Additional marking. In addition to the above marking, the following information shall be legibly printed on both sides of each appropriate pouch:

Trichloromelamine pouch A
Potassium iodide pouch B

3.5.3 Instruction sheet. A sheet of chemical wood writing paper, legibly printed with the instructions seen on figure 3, shall be provided as specified in the acquisition order (see 6.2).

3.6 Workmanship. The food service disinfectant shall be a noncaking, clean, uniform powder. The entire process of mixing, grinding (see 6.3, 6.4, and 6.5), filling, and sealing the individual pouches shall be carried out in surroundings maintained at a relative humidity of not more than 45 percent, at a temperature of not more than 80 ºF, and in such a manner that ensures a product of consistently high quality. All components and bulk mixed finished formulation shall be unit packed in moisture-vapor proof containers and kept dry and protected from moisture until incorporated into final packaging.

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3.6.1 Additional workmanship. The potassium iodide cannot be mixed intimately with the rest of the pouch A formulation. The unit packing of the potassium iodide and the pouch A finished product shall be carried out in separate rooms under the controlled humidity and temperature conditions specified in 3.6. A double-compartmented pouch type package shall be used to keep the potassium iodide separated from the pouch A formulation until time of use.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Surface active material (LAS) inspection (see 4.2).
b. In-process inspection (see 4.3).
c. End item conformance inspection (see 4.4).

4.2 Surface active material inspection (LAS). Surface active material testing shall be performed in accordance with table II. Each lot of surface active material shall be inspected and meet all requirements prior to incorporation into the formulation. For purposes of sampling, a lot shall consist of all the surface active material received from one manufacturer that was produced at one plant from the same raw materials under the same manufacturing conditions. The lot size shall be expressed in pounds of surface active material. The sample unit for testing, except color, shall be a 1/2-pound composite obtained by randomly selecting and thoroughly mixing 1/4-pound samples in the frequency indicated below. The composite shall be placed in a clean, dry, sealed container of the smallest possible size. Care shall be exercised to prevent contamination or alteration of the composite during the sampling, weighing, and storage. The lot shall be unacceptable if the composite fails to meet any of the test requirements specified. All test reports shall contain the individual values utilized in expressing the final result.

Lot size (pounds)	Frequency
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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TABLE II. Instructions for testing of surface active material.

Characteristic	Document reference		Requirement applied to		Number of determinations per unit	Results reported as	
	Requirement	Test method	Ind. unit	Lot avg.		Pass or fail	Numerical value
Surface active material color	3.1.2	3.1.2	X[1]			X	
Iodine uptake	3.1.2.1	4.5.1.1		X	2		Average of 2 determinations to nearest 0.5 mg
Active ingredient	3.1.2.2	4.5.1.2		X	2		Average of 2 determinations to nearest percent
Moisture	3.1.2.3	4.5.1.3		X	2		Average of 2 determinations to nearest 0.01 percent
Potassium iodide	3.1.3	[2]					

[1]All containers shall be examined for color. Failure of the contents of a container to meet the color requirement shall be cause for rejection of the contents of that container.
[2]The contractor shall furnish a certificate of compliance covering this requirement.

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4.3 In-process inspection. Inspection shall be made of the following operations or assemblies to establish conformance to specified requirements. Whenever non-conformance is noted, correction shall be made to the items affected and to the operation. Items that cannot be corrected shall be removed from production.

a. Batch mix composition formulation complies with 3.2.
b. Workmanship complies with 3.6.

4.4 End item conformance inspection. Inspection shall be made of the end item for the requirements specified in 4.4.1 and 4.4.2.

4.4.1 End item visual examination. The end item shall be examined for the defects listed in table III. The lot size shall be expressed in units of filled bags. The sample unit shall be one filled bag.

TABLE III. End item visual defects.

Examine	Defects	Classification	
		Major	Minor
Construction	Outer bag not constructed of materials specified.		201
	Outer bag not sealed.	101	
	Pouch A (trichloromelamine) missing.	102	
	Pouch B (potassium iodide) missing.	103	
	Tearing notch missing from pouch.		202
	Tearing notch not positioned as specified.		203
	Seams and closures of pouches not completely heat sealed throughout entire length and width.	104	
	Not uniformly constructed.		204
	Pouches not constructed of materials specified.		205
	Any evidence of seepage; any unsound seam.	105	
	Any tear, cut, slit, split, cracks, blisters, or holes (including pinholes).	106	
Workmanship	Not clean; any evidence of dirt, dust, or foreign matter.	107	
	Presence of food service disinfectant or potassium iodide on exterior of pouches, as applicable.		206
	Caking; powder not uniform.	108	

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TABLE III. End item visual defects - Continued.

Examine	Defects	Classification	
		Major	Minor
Markings	Markings pertaining to the applicable information and instructions for end item use missing, illegible, incomplete or do not conform to applicable legend specified in 3.5.	109	
	Not printed in black or blue ink.		207
	Printing ink faded, blurred, or smeared.		208

4.4.1.1 End item dimensional examination. The filled bags shall be examined for the defects listed in table IV. The lot size shall be expressed in units of filled bags. The sample unit shall be one filled bag.

TABLE IV. End item dimensional defects.

Examination of pouch seam size	Defects	Classification	
		Major	Minor
Two piece construction	Side, top, and bottom seams less than 3/16 inch wide.		209
One piece construction	Side seams and closure less than 3/16 inch wide.		210
	Bottom fold seal less than 3/16 inch wide.		211
	Less than 3/8 inch lip present for joining and sealing pouch A and pouch B.		212

4.4.1.2 Examination of the filled pouches for net contents. The lot size shall be expressed in units of filled pouches. The sample unit shall be one filled pouch. The net contents of pouch A and pouch B shall not be less than as specified in 3.3.

4.4.2 End item testing of pouch A. The methods of testing listed in table V shall be followed. The lot size shall be expressed in units of bags. The sample unit for testing the pouch A contents shall be five individual 3.43-ounce pouches. The number of sample units to be tested shall be as indicated below. Tests shall be performed on a sample unit or composite basis in accordance with table V. The lot shall be unacceptable if a sample unit or the composite fails to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final results.

Lot size (bags)	Number of sample units
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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4.4.2.1 Composite sample. One 3.43-ounce pouch A shall be selected from each sample unit. The contents of the pouches selected shall be thoroughly mixed to yield a homogeneous composite. The composite shall be placed in an airtight and watertight brown glass container and covered and sealed to prevent any moisture change.

4.4.2.2 Sample unit basis. The remaining pouches from the sample units selected shall be used for testing on a sample unit basis. Care shall be exercised to prevent any change in the compound due to moisture prior to testing.

4.4.2.3 Frequency of testing. Except when otherwise specified in table V, the tests for each characteristic shall be performed on a lot-by-lot basis. Those tests designated by an asterisk shall be performed on the first lot and on every fifth lot thereafter.

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TABLE V. Instructions for testing of the end item.

Characteristic	Document reference		Requirement applied to		Number of determinations per unit	Results reported as	
	Requirement	Test method	Ind. unit	Lot avg.		Pass or fail	Numerical value
Composition	Table I or Table II	[1]					
Contents of pouch A:							
Titratable chlorine	3.4.1	4.5.2.1	X		2		Average of 2 determinations to nearest 0.1 percent
Acidity:	3.4.2						
pH value		4.5.2.2	X		1		0.1 pH
Titratable acidity		4.5.2.2	X		1		mg
Surface active ingredient	3.4.3	4.5.2.3		X	3		Average of 3 determinations to nearest 0.1 percent
Solubility	3.4.4	4.5.2.4		X	1	X	mg
Moisture	3.4.5	4.5.2.5	X		2		Average of 2 determinations to nearest 0.01 percent
Particle size	3.4.6						
Standard sieve No. 20		4.5.2.6		X	1		0.1 percent/weight
Standard sieve No. 10		4.5.2.6		X	1		0.1 percent/weight
Bactericidal performance[2]	3.4.7	4.5.2.7		X	3		Average of 3 determinations to 0.001 percent kill

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TABLE V. Instructions for testing of the end item - Continued.

Characteristic	Document reference		Requirement applied to		Number of determinations per unit	Results reported as	
	Requirement	Test method	Ind. unit	Lot avg.		Pass or fail	Numerical value
Formulation stability	3.4.8	4.5.2.8	X		1	X	
Surface tension	3.4.9	4.5.2.9		X	3		Average of 3 determinations to nearest dyne/cm
Pouches, filled and sealed:							
Finished pouch stability	3.3.4	4.5.3.1	X		2	X	
Leakage of seams and closures	3.3.5	4.5.3.2	X		1	X	

[1]Certificate of compliance shall be furnished by the contracting officer for composition requirements except for surface active material.
[2]Tests performed on the first lot and/or every fifth lot thereafter.

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4.5 Methods of inspection.

4.5.1 Tests on surface active material.

4.5.1.1 Iodine uptake. Weigh a sample of 2 grams of the LAS surface active material to the nearest 0.005 gram on a piece of waxed paper. Transfer it to a 250-mL separatory funnel (Corning No. 6340 or equivalent with stem cut to approximately 0.5 inch length) containing 50 mL of hot distilled water. Dilute to 75 mL with hot distilled water and swirl but do not shake to dissolve. Add 2 mL of starch indicator solution and titrate with 0.1 normal (N) iodine solution (12.692 grams iodine and 8 grams potassium iodide per liter of distilled water) until all sulfate is destroyed as indicated by the appearance of the starch-iodine color. Swirl during the titration and allow to stand with occasional swirling for 2 minutes after the initial appearance of the starch-iodine color to be certain that the color does not fade.

4.5.1.1.1 Calculation. Calculate the result as follows:

$$\text{Milligrams iodine} = \frac{(126.92)(\text{mL of iodine solution})(N)}{(\text{grams of sample})}$$

4.5.1.2 Active ingredient. Weigh out 2 grams of LAS to the nearest 0.1 mg in a plastic dish. Transfer to a 250-mL volumetric flask that is half full with distilled water. Mix well by shaking and dilute to volume 250 mL with distilled water. Then mix again. Add 25 mL of methylene blue indicator, 25 mL chloroform, 25 mL distilled water, and pipette a 25-mL aliquot of LAS slowly into a 250-mL graduated cylinder. From a 50-mL burette, add as large a volume as possible of 0.018 N hyamine 1622 without risk of over titrating. Stopper the graduated cylinder and shake at least 30 times. To speed separation, hold the graduated cylinder in a horizontal position until most of the chloroform has separated, then return the graduated cylinder to the vertical position and swirl gently. After the mixture has separated, note color difference between the two layers. Using the degree of color difference as a guide, continue adding hyamine in successively smaller increments, shaking the graduated cylinder 30 times after each addition. When colors are close, add the hyamine in 0.05-mL amounts until end point, at which both will be aqua blue color. The end point is when the colors of the two layers match when the graduated cylinder is observed at eye level against a white background. The comparison should be made as soon as the layers separate completely.

4.5.1.2.1 Calculation. Calculate the result as follows:

$$\text{Percent active ingredient} = \frac{(\text{mL hyamine})(N \text{ hyamine})(\text{M.W.})(\text{dilution})}{(10)(\text{sample weight})(\text{aliquot})}$$

Where:

N = Normality of hyamine solution (see sections 4.5.1.2.5 and 4.5.1.2.6)
M.W. = Molecular weight of LAS (=340)
Dilution = Volume LAS made to (250 mL) of LAS solution
Aliquot = Volume of LAS solution used for titration (25 mL)

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4.5.1.2.2 Reagents - hyamine 1662.

a. Hyamine 1622 crystals (benzethonium chloride).
b. Sodium hydroxide, 50 percent solution, reagent grade.
c. Sodium lauryl sulfate (dodecyl sulfate, sodium salt) crystals.

4.5.1.2.3 Reagents - methylene blue indicator.

a. Methylene blue chloride (652).
b. Sulfuric acid, concentrated, reagent grade.
c. Sodium sulfate, anhydrous, reagent grade.

4.5.1.2.4 Preparation of hyamine 0.018 N. Fill a one-liter volumetric flask half full with distilled water. Then carefully add 3.20 grams of 50 percent sodium hydroxide and mix. Then add 7.36 grams of hyamine crystals and fill to volume with distilled water. Stir until dissolved. Do not dry the hyamine. The weight should be adjusted, if necessary, to provide a solution close to 0.018 N.

4.5.1.2.5 Standardization of hyamine. Sodium lauryl sulfate (SLS) active solution; (by weight), 0.003 N: weigh approximately 0.873 grams of SLS to the nearest 0.1 mg (use standard as received, do not dry). Transfer to a one-liter volumetric flask, dissolve in distilled water, dilute to volume, and mix. Calculate the normality of SLS solution to six decimal places. When not being used, store the SLS in a tightly stoppered container. Label with normality, date and technician's initials. Discard after one year. To standardize hyamine, pipette a 20-mL aliquot from the 0.003 N standard active SLS solution and add it to a 250-mL stoppered graduated cylinder already containing:

25 mL methylene blue indicator
25 mL chloroform
30 mL distilled water

Titrate with the 0.002 N hyamine solution using a 50-mL burette containing as large a volume as possible without risk of over-titrating. Stopper the graduated cylinder and shake vigorously at least 30 times. To speed up the separation, hold the graduated cylinder in a horizontal position until most of the chloroform has separated, then return the graduated cylinder to the vertical position and swirl gently. After the mixture has separated, note color difference between the two layers. Using the degree of color difference as a guide, continue adding hyamine in successively smaller increments, shaking the graduated cylinder 20 times after each addition. When colors of the two layers are close, add the hyamine in 0.5-mL amounts until the end point at which both layers will be an aqua blue color. The end point is when the colors of the two match when the graduated cylinder is observed at eye level against a white background. The comparison should be made as soon as the layers separate completely.

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4.5.1.2.6 Calculation. Calculate the result as follows:

$$\text{Normality of sodium lauryl sulfate (N SLS)} = \frac{(\text{weight of SLS})(\text{percent purity})}{(100)(288.4)}$$

Where:

$$\text{N hyamine} = \frac{20 \text{ X (N SLS)}}{\text{mL hyamine}}$$

4.5.1.2.7 Labeling. Label the container with the normality, date, and technician's initials.

4.5.1.2.8 Caution. Hyamine normality is affected by contact with certain types of plastic tubing. Therefore, it may be necessary to flush the tubing if solution has been standing two days or more.

4.5.1.2.9 Preparation of methylene blue indicator. Prepare a dye solution containing 3.0 grams of pure methylene blue per liter. The amount of dye depends on the percent dye stated on the container label and is calculated as follows:

$$\text{Grams of dye} = \frac{3.00 \text{ X percent dye content}}{100}$$

Wash the dye into a one-liter volumetric flask and dilute to volume with distilled water. Mix well before each use. To prepare the methyl blue indicator, fill the container to be used half full with distilled water. Then for each liter of solution add 10 mL of methylene blue concentrate, 8 mL of concentrated sulfuric acid, and 50 grams of sodium sulfate. Mix well and dilute to volume. Mix well again. Label properly.

4.5.1.3 Moisture. Weigh sample of 3 grams of the LAS surface active material to the nearest 0.1 mg in a covered sample dish. Uncover the dish and dry the sample in an oven for 24 hours at 105 ºC ± 2 ºC. Replace the cover, cool the sample to room temperature in a desiccator, and again weigh the sample to the nearest 0.1 mg.

4.5.1.3.1 Calculation. Calculate the result as follows:

$$\text{Percent moisture} = \frac{(\text{loss of weight})(100)}{(\text{grams of sample})}$$

4.5.2 Tests on contents of pouch A.

4.5.2.1 Titratable chlorine. Thoroughly and quickly mix approximately 5 grams of the contents of the pouch A sampled at random by shaking in a closed glass container. Promptly obtain a representative sample of 1 gram from the 5 gram sample, weighed to the nearest milligram, and place in a 500-mL Erlenmeyer flask. To this, add 2 grams of potassium iodide (KI) crystals and approximately 25 mL of glacial acetic acid. Stopper the flask and swirl for about 5 minutes to dissolve the sample. (Note: Carry out determination of titratable chlorine in

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subdued light.) Add approximately 25 mL of distilled water and again stopper and swirl to ensure complete mixing and solution of KI. Titrate at once with a freshly standardized 0.15 N sodium thiosulfate solution to light straw color. Add starch indicator, and continue titration to end point.

4.5.2.1.1 Calculation. Calculate the result as follows:

$$\text{Percent titratable chlorine} = \frac{(35.45)(\text{mL of sodium thiosulfate})(N)(100)}{(\text{grams of sample})(1000)}$$

4.5.2.2 Acidity. Weigh 1 gram of the contents of the pouch A to the nearest milligram and disperse in 1 liter of distilled water at 25 ºC ± 1 ºC. The pH, measured with a glass electrode, shall not be lower than 2.5. Titrate the entire solution with 0.1 N NaOH. The latter solution is added from a burette while stirring, measuring the pH with a glass electrode at suitable increments. As the pH value approaches 5.0, the increments should be made smaller, with pH intervals no greater than 0.2. Continue the titration until pH of 5.5 has been reached. The number of milliliters of 0.1 N NaOH required to give a pH of 5.0 is then found by interpolation of this data.

4.5.2.2.1 Calculation. Calculate the result as follows:

$$\text{Milligrams NaOH per gram} = \frac{(\text{mL of NaOH})(N)(40)}{(\text{grams of sample})}$$

Where:
N = Normality of NaOH

4.5.2.3 Surface active ingredient. A 1.5-gram sample of disinfectant from pouch A weighed out to the nearest mg shall be dispersed in a 250-mL volumetric flask containing 125 mL of distilled water. Mix well and fill to volume. Mix well again. Pipette a 25-mL aliquot of sample into a 125-mL flask. Add 2.0 N sodium arsenite drop wise. Test one drop of solution on starch iodine test paper until all blue disappears (available chlorine is removed). Record number of drops required. Discard. To a 250-mL stoppered graduated cylinder add 25 mL of methylene blue indicator, 25 mL chloroform, 25 mL of distilled water, and a 25-mL portion of the sample taken from the original flask while agitating. Add predetermined volume of sodium arsenite plus an excess of 1 or 2 drops as determined in the pretest. Titrate with standardized hyamine 1622 solution (0.002 N). Shake vigorously after each addition. After the mixture has separated, note color difference between the two layers. Using the degree of difference as a guide, continue adding hyamine in successfully smaller increments, shaking the graduated cylinder two times after each addition. When colors are close, add the hyamine in 0.05-mL amounts until the end point.

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4.5.2.3.1 Calculation. Calculate the result as follows:

$$\text{Percent surface active ingredient} = \frac{(\text{mL hyamine})(\text{N hyamine})(\text{M.W.})(\text{dilution})}{(10)(\text{sample weight})(\text{aliquot})}$$

Where:
N = Normality of hyamine solution (see sections 4.5.1.2.5 and 4.5.1.2.6)
M.W. = Molecular weight of LAS (=340)
Dilution = Volume of sample (from pouch A) made up to (250 mL)
Aliquot = Volume sample solution used for titration (25 mL)

4.5.2.3.2 Preparation of hyamine 1622 (0.002 N). Fill a one-liter volumetric flask half full with distilled water. Add 0.4 grams 50 percent NaOH slowly and mix. Then add 0.82 grams of hyamine crystals. Stir until dissolved. Fill to volume with distilled water. For rest of standardizing procedures, see 4.5.1.2.5.

4.5.2.3.3 Preparation of methylene blue. For preparation of methylene blue, see 4.5.1.2.9.

4.5.2.3.3.1 Reagents.

a. Sodium arsenite (2N) $NaAsO_2$, reagent grade.
b. Sodium hydroxide solution 50 percent, reagent grade.

4.5.2.3.4 Procedure. Wash 32.5 grams of sodium arsenite into a 250-mL volumetric flask, with distilled water. Add 19.7 mL of 50 percent sodium hydroxide, dilute to volume with distilled water, and mix well to give 2 N solution. The solution is highly alkaline and susceptible to air oxidation. It should be stored in a plastic container and dispensed from a dropping bottle. NOTE: Promptly identify solution with a POISON label.

4.5.2.4 Solubility. Measure one liter of distilled water at 25 ºC ± 1 ºC into a 1.5-liter beaker. Weigh out 1 gram of the contents of the pouch A to the nearest milligram. Add carefully to the distilled water and stir for 2 minutes with a glass rod. Stirring should not be so rapid as to cause splashing or whipping of the mixture but sufficient to keep the undissolved powder from settling to the bottom. Withdraw the stirring rod and allow the solution to stand 2 minutes. At the end of this time, insert a 20-mL pipette about 1 inch below the surface and draw 20 mL into the pipette. Place the 20-mL sample in a 125-mL Erlenmeyer flask and analyze for titratable chlorine by the following procedure: Dissolve 1 gram of KI crystals in the 20-mL sample. Add 6 drops of 50 percent acetic acid and swirl to ensure complete mixing. Allow to stand 5 minutes in the dark. Titrate the free iodine at once with freshly standardized 0.005 N sodium thiosulfate. When the color is straw yellow, add starch indicator and titrate to end point.

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4.5.2.4.1 Calculation. Calculate the result as follows:

$$\text{Milligrams of titratable chlorine per liter} = \frac{(35.45)(\text{mL sodium thiosulfate})(N)(1000)}{(\text{mL of sample solution})}$$

Where:
N = Normality of thiosulfate solution

4.5.2.5 Moisture. Weigh to the nearest 0.1 mg a sample of 3 grams of the contents of the pouch A in a covered sample dish. Uncover the dish and dry the sample in a vacuum oven for 2 hours at 50 ºC ± 2 ºC. Replace the cover, cool the sample to room temperature in a desiccator, and again weigh the sample to the nearest 0.1 mg.

4.5.2.5.1 Calculation. Calculate the result as follows:

$$\text{Percent moisture} = \frac{(\text{loss of weight})(100)}{(\text{grams of sample})}$$

4.5.2.6 Particle size. Weigh a representative 10 gram ± 0.05 gram sample of the contents of the pouch A and place on a No. 10 standard sieve that is fitted into a No. 20 standard sieve, both sieves conforming to ASTM E 11. The No. 20 sieve is fitted on the receiving pan. After introducing the sample, place a top on the No. 10 sieve. Gently tap the sieve assembly until no further material will pass through either sieve, as determined by observation. Weigh the material that was collected in the bottom pan to the nearest 0.05 gram. The weight of the sample shall be not less than 9.5 grams.

4.5.2.6.1 Calculation. Calculate the result as follows:

$$\text{Percent passing No. 20 sieve} = (\text{grams of material in pan})(10)$$

4.5.2.7 Bactericidal performance. The following procedure should be carried out on three separate 100-mg portions of the contents of the pouch A to give three independent sets of results:

a. Aseptically transfer to four sterilized 250-mL Erlenmeyer flasks 93 mL of cooled, sterilized hard water prepared as follows: To one liter of sterilized, cooled, glass distilled water, add 0.54 grams of sodium bicarbonate and 0.36 grams of anhydrous calcium chloride. CAUTION: Add these ingredients aseptically to the cooled sterilized water; hot water may cause decomposition of the bicarbonate.

b. Mark the flasks 1, 2, 3, and 4 and place in a 45 ºC ± 1 ºC constant temperature water bath for 30 minutes or more. Weigh three 100-mg samples of the finished product of the pouch A to the nearest 0.1 mg and introduce them separately into flasks 1, 2, and 3. Swirl the contents to facilitate solution and return the flasks to the bath.

c. After 10 minutes, pipette 2 mL of a freshly made sterile 10 percent Difco™ Bacto® Peptone solution or equal into each of the four flasks. Swirl the contents again and return to the bath.

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d. After 120 minutes, pipette 5 mL of a suspension of bacterium Escherichia coli strain No. 26 ATCC (see 4.5.2.7.1) containing approximately two billion cells per milliliter into each of the four flasks using sterile pipettes and mix thoroughly by continuous swirling in the bath.

e. After a contact of 90 seconds, pipette 2 mL of the final mixture in each of flasks 1, 2, and 3 separately into three sterile test tubes each containing 2 mL of a sterile 1 percent sodium sulfite (Na_2SO_3) solution (made up to contain 1.0 percent Na_2SO_3, 0.1 percent sodium pyro phosphate ($Na_4P_2O_71OH_2O$) and 0.2 percent glycerine that is prepared no more than 24 hours before use. (NOTE: All glassware used in contact with the test fluid must be sterilized and a fresh pipette must be used for each sample. Both the bacteria-dosing and the taking of sample must be timed so that the bacteria in each of the three test flasks shall have exactly 90 seconds of contact with the test fluid.)

f. Pour triplicate plates with 1-mL portions of each of the three neutralized samples before counting using tryptone glucose extract agar. Remove a 1-mL portion from flask 4 into 99 mL sterile water and mix thoroughly using tryptone glucose extract agar. Make two more 1:100 serial dilutions in the same manner and pour triplicate plates with 0.1- and 1.0-mL portions of the $1:10^6$ dilution using tryptone glucose extract agar and incubate at 37 ºC ± 1 ºC for 48 hours before counting. Multiply by two the counts on plates poured with the three test samples to correct for the dilution. Compute the percentage of survivors in each of the three test flasks from the average count per milliliter of each of the test samples and that of the control sample from flask 4.

4.5.2.7.1 Inoculum. Wash off the surface growth on several 18-hour old nutrient-agar slants with sterile, normal saline, pooled and filtered through sterile No. 2 Whatman filter paper. Adjust the suspension to 2.0 billion cells per milliliter with a photoelectric colorimeter or other suitable means. Store in the refrigerator until used. The suspension must be prepared fresh each day.

4.5.2.8 Formulation stability. Seal triplicate samples of the contents of the pouch A of approximately 50 grams each (the titratable chlorine having been determined by the method specified in 4.5.2.1) in 4-ounce screw cap glass jars or other suitable glass containers and store for 60 days ± 4 hours in a suitable constant temperature oven maintained at 50 ºC ± 1 ºC. At the end of the test period, determine the titratable chlorine of each sample by the method specified in 4.5.2.1. The average of the three samples shall be taken as the final value of titratable chlorine.

4.5.2.8.1 Calculation. Calculate the result as follows:

$$\text{Percent titratable chlorine remaining} = \frac{(\text{final value})(100)}{(\text{original value})}$$

4.5.2.9 Surface tension. Grind and mix thoroughly approximately 5 grams of the contents of the pouch A in a glass mortar with a glass pestle. Weigh a 1-gram sample that has been obtained by sampling various portions of this finely ground and well mixed material to the nearest 0.1 mg, and dissolve in distilled water. Make up to 1 liter volume. Three suitable portions of this solution shall be used for determination of surface tension at 25 ºC using a calibrated du Nouty tensiometer or a Roller Smith surface tension balance (platinum ring method). The average of the three determinations is obtained and expressed as dynes per centimeter.

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4.5.3 Tests on filled and sealed pouches.

4.5.3.1 Finished pouch stability. Place the finished pouch or pouches, as applicable, (filled and heat sealed) in a humidity cabinet for a period of 15 days ± 1 hour. Maintain the temperature and humidity range of the test cabinet at 38 ºC ± 1 ºC with a minimum relative humidity of 90 percent.

4.5.3.2 Leakage of seams and closures of pouch. Place one filled and heat sealed pouch in a desiccator or other suitable container partially filled with water. Cover the pouch with a grating to prevent floating and to maintain a one-inch height of water over the top surface of the filled pouch. Gradually apply a vacuum of 20 inches of mercury for a period of 3 minutes. Observe the pouch for a steady flow of air bubbles (leakage indicating) during this period. Isolated or static bubbles caused by entrapped air in the seams or on the surface of the pouch will not be considered a sign of leakage. Remove the pouch from the desiccator or other suitable container, dry hands and pouch, and examine the inside of the pouch for presence of moisture.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the inventory control point's packaging activity within the military department or defense agency, or within the military department's system command. Packaging data retrieval is available from the managing military department's or defense agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 DOT class and shipping name. Food service disinfectant produced in accordance with this specification shall be marked and labeled in accordance with DOT class 8 requirements. The applicable shipping name is "Corrosive Solids, N.O.S. (Trichloromelamine), 8, UN1759, III.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The food service disinfectant is intended for use in disinfection of fresh fruits and vegetables grown in areas where use of feces as fertilizer is suspected. The food service disinfectant is also intended for use in disinfection of mess gear and eating utensils. The food service disinfectant provides health protection to personnel against diseases such as amoebic dysentery, and under emergency conditions, it constitutes their sole protection. It is intended for use by troops and the amount packaged is sufficient for treating 25 gallons of water and 100 mess kits. The food service disinfectant covered by this document is intended for use only when hot water rinse is not available.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).
c. When weight of pouch required differs from specified weight (see 3.3).
d. Instruction sheet for determination of titratable chlorine (see 3.5.3).
e. Packaging (see 5.1).

6.3 Component handling. Extreme care must be exercised in handling, grinding, mixing and packaging the components and formulation to protect against moisture and to ensure homogeneity of the formulation so that a consistently high quality product is produced that will remain stable during long time storage. All equipment should be kept clean and protected against moisture during any shutdown. During a shutdown of more than 12 hours, it is recommended that all adhering formulation be removed from the grinding, mixing, and filling machines.

6.4 Specialized requirements. Contractor's attention is directed to the fact that this item is difficult to produce and requires specialized equipment and technical knowledge based on past experience to produce the superior product required. There can be no compromise or deviation from the high quality and stability standards set for this product. Components for surface active material and potassium iodide having the properties listed below have been used successfully in the past to prepare a food service disinfectant that conforms to this document.

6.4.1 Trichloromelamine. The trichloromelamine should be white to light cream in color, having a titratable chlorine of not less than 89.0 percent and no more than 94.5 percent, and a moisture content not greater than 0.5 percent.

6.4.1.1 Citric acid, anhydrous. The anhydrous citric acid should have a moisture content of not greater than 0.2 percent and the assay show not less than 99.3 percent $C_6H_8O_7$ should conform to the requirements of the USP-NF.

6.4.1.2 Monosodium dihydrogen phosphate, anhydrous. The anhydrous monosodium dihydrogen phosphate should contain not less than 57.5 percent nor more than 59.1 percent of phosphorous pentoxide and should have a moisture content no greater than 2.0 percent.

6.5 Grinding. In order to meet solubility requirements consistently, the procedures listed in 6.5.1 and 6.5.2 should be followed.

6.5.1 Contents of trichloromelamine pouch. It is necessary to grind at least a portion of the finished formulation and all of the trichloromelamine and wetting agent in micronizing equipment. Trichloromelamine should not be ground alone or when mixed only with the monosodium dihydrogen phosphate because of fire hazard.

6.5.2 Contents of chlorinated cyanurate. It is necessary to grind all of the ingredients other than chlorinated cyanurate in micronizing equipment. Powder or regular grade potassium dichloro-s-triazinetrione should be added in the end when dry mixing all the ingredients in the formulation.

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6.6 Laboratory report approval. At the time a bid is submitted, the contractor should submit to the contracting officer a certified copy of a current year laboratory report covering the quality and grade of materials that the contractor proposes to deliver. The laboratory report should contain test data demonstrating that the food service disinfectant proposed for delivery by the contractor has been tested in accordance with section 4 and found to comply with the requirements of this document. Any one of the following types of reports will be satisfactory for this requirement:

a. An independent or commercial laboratory report.
b. The prospective contractor's own laboratory report.
c. A government laboratory report from a contract.

The purpose of this requirement is to assist in determining the capability of contractors to conform to all requirements of this document. The submission of an acceptable report under this requirement should not be construed as relieving a contractor from subsequently meeting all requirements of this document on all deliverables.

6.7 Subject term (key word) listing.

bactericidal
chlorine
citric acid
iodine
monosodium dihydrogen phosphate
potassium iodide
trichloromelamine

6.8 Changes from the previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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DISINFECTANT, FOOD SERVICE

(CHLORINE-IODINE TYPE)
6840-00-810-6396
MIL-DTL-11309J-TYPE I

For disinfection of mess gear, fresh fruits, and vegetables
(4.77 oz. Net quantity for 25 gallons of water.)

ACTIVE INGREDIENTS:
Trichloromelamine ..... 19.3 %
Potassium Iodide ..... 28.6 %
INERT INGREDIENTS: ..... 52.1 %
TOTAL ..... 100.0 %

AVAILABLE CHLORINE	250 mg/l (ppm)
TITRATABLE IODINE	300 mg/l (ppm)

Keep Out of Reach of Children

DANGER

SEE BACK PANEL FOR ADDITIONAL PRECAUTIONS AND FIRST AID

Manufactured For:
US Army Research, Development, and
Engineering Command
Natick Soldier Center
AMSRD-NSC-CF-F
15 Kansas St.
Natick, MA 01760-5018

Lot No. Date

Contract Holder:

[Name & Address]

Contract No.

EPA Est. No.

EPA Reg. No. 40510-1

Inspection Test Date:

FIGURE 1. EPA approved front label.

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DIRECTIONS FOR USE

It is a violation of Federal Law to use this product in a manner inconsistent with its labeling.

a. For disinfection of mess gear (intended for use only when hot water rinse is not available):
The double compartmented pouch is sufficient for treating 25 gallons of water resulting in a dosage of approximately 257 mg/l of chlorine and 300 mg/l of iodine. To treat 100 sets of mess gear, dissolve the contents of one double compartmented pouch of Disinfectant, Food Service (consisting of one trichloromelamine pouch labeled pouch A and one potassium iodide pouch labeled pouch B) in a container containing 25 gallons of water. Repeat the above procedure with a second container containing 25 gallons of rinse water. If water is cold, mix contents of the double compartmented pouch with 3 to 4 parts of water in a small container (canteen cup will do) and pour the mixture into the rinse water. Repeat for the second container of 25 gallons of water. Stir thoroughly to dissolve. After washing each set of mess gear, rinse twice - once in the first container of disinfectant solution and a final rinse in the second container of disinfectant solution. Allow mess gear to air dry. A new solution must be prepared after disinfecting 100 mess kits.
b. For fresh fruits and vegetables:
1. Remove bruised and torn leaves and damaged fruits or vegetables.
2. Do not cut or peel fruits or vegetables before disinfecting.
3. Wash thoroughly in a solution made by dissolving the contents of one double-compartmented pouch (pouch A and pouch B) in 20 gallons of water. This solution will yield approximately 320 mg/l of chlorine and 380 mg/l of iodine.
4. After washing, completely immerse the fruits and vegetables for ten (10) minutes in a separate freshly prepared solution made by dissolving one double-compartmented pouch of Disinfectant, Food Service in 20 gallons of water. Stir occasionally to obtain thorough wetting of vegetables and fruit surfaces.
5. After ten (10) minutes of immersion, remove the fruits and vegetables from the disinfectant solution and rinse thoroughly in potable water.
NOTE: SOLUTIONS OF DISINFECTANT, FOOD SERVICE ARE NOT TO BE REUSED, MAKE FRESH SOLUTIONS FOR EACH USE

STORAGE AND DISPOSAL

Store in a cool, dry, well ventilated area away from contact with other chemicals and solvents. Moisture contamination may liberate irritating and hazardous chlorine containing gases. Rinse empty pouches thoroughly with water and discard them in a safe place.

PRECAUTIONARY STATEMENTS
HAZARD TO HUMANS AND DOMESTIC ANIMALS

Harmful if swallowed. May cause eye damage. Do not get in eyes, on skin or clothing. Do not breathe dust.

FIRST AID:

IF IN EYES: Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye. Get immediate medical attention.
IF ON SKIN OR CLOTHING: Take off contaminated clothing. Rinse skin immediately with plenty of water for 15-20 minutes. Get immediate medical attention.
IF INHALED: Move person to fresh air. If person is not breathing, call medical personnel, then give artificial respiration, preferably by mouth-to-mouth if possible. Get immediate medical attention.
IF SWALLOWED: Have person sip a glass of water if able to swallow. Do not induce vomiting unless told to do so by medical personnel. Do not give anything by mouth to an unconscious person. Get immediate medical attention.

Note to Physician: Probable mucosal damage may contraindicate the use of gastric lavage.

PHYSICAL OR CHEMICAL HAZARDS

Contains trichloromelamine which may react vigorously to evolve heat and highly toxic smoke under the following conditions: If it is exposed to extreme heat, if ignited by spark or flame, if mixed before dilution in water with certain organic materials, acid and alkalis.

DO NOT MIX WITH OTHER CHEMICALS SUCH AS CLEANING COMPOUNDS,AMMONIA, ACIDS, SOLVENTS, OR OTHER ORGANIC MATERIAL.

FIGURE 2. EPA approved back label.

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INSTRUCTION FOR DETERMINATION OF TITRATABLE CHLORINE WHEN DISINFECTANT, FOOD SERVICE IS TO BE ISSUED AFTER _____ (the same date stamped on caution label affixed to shipping container shall be inserted here).

Weigh out 1 gram of the contents of the pouch A to the nearest milligram into a round bottom test tube (approximately 14 millimeters (mm) in diameter and 160 mm in length). Add 0.5 milliliters (mL) of distilled water to the sample, and stir with a round tipped stirring rod (of a size convenient to fit into tube). Mix the sample into a smooth paste for exactly 3 minutes. Completely transfer the paste by rinsing into a 1.5 liter beaker, using a total of exactly 1 liter of distilled water. Stir the solution for exactly 30 seconds with a large stirring rod (do not whip or splash when stirring) and then let stand for 2 minutes. At the end of this time, insert a 20-mL pipette 1 inch below the surface and draw 20 mL into the pipette. Place the 20-mL sample into a 125-mL Erlenmeyer flask or iodine flask and analyze for titratable chlorine by the following procedure:

Dissolve 1 gram of potassium iodide crystals into the 20-mL sample. Add six drops of 50 percent acetic acid and swirl to ensure complete mixing. Allow to stand five minutes in the dark. Titrate the free iodine at once with freshly standardized 0.005 N sodium thiosulfate. When the color is straw yellow, add starch indicator and titrate to end point. Analysis should be conducted in triplicate and results reported as an average. Calculate the result as follows:

$$\text{Percent titratable chlorine} = \frac{(35.45)(\text{mL of sodium thiosulfate})(N)(50)(100)}{(\text{gram sample})(1000)}$$

Where: N = normality of sodium thiosulfate solution.

FIGURE 3. Instruction for determination of titratable chlorine.

Custodians:
Army - GL
Navy - MC
Air Force - 68

Review Activities:
Army - MD
Navy - MS

Preparing Activity:
DLA - GS3

(Project 6840-2005-001)

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